



LIANGSHAN ENTERPRISE

RISK ASSESSMENT

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iii. Risk Assessment

1. Asset CIA Classification

Note: Submitted the excel sheet separately for better viewing.

S. No.	Asset Name (with Value)	Value				Asset Classification (H, M, L)	Remarks
		Confidentiality©	Integrity(I)	Availability(A)	Score (C* I* A)		
1	Proprietary BIM Models	3	3	3	27	High	Critical for design execution and innovation; protected with RBAC and encryption.
2	Client Contracts	3	3	2	18	High	Confidential agreements; encrypted and access restricted.
3	Research Data	3	3	2	18	High	Proprietary research on sustainable materials; highly sensitive.
4	Financial Records	3	3	3	27	High	Core to operations; backed up securely and access limited to authorized personnel.
5	Employee Data/Records	3	2	2	12	Medium	Personal data; complies with GDPR and other privacy laws.
6	IT Systems	2	3	3	18	High	High availability required for global collaboration; redundant systems in place.
7	IoT Devices	2	2	2	8	Medium	Used for smart building management; requires regular firmware updates to mitigate vulnerabilities.
8	Data Centers	3	3	3	27	High	Store mission-critical data; ISO 27001-compliant.

Table 3.1: Asset Classification Table

2. Asset Inventory Information

Note: Submitted the excel sheet separately for better viewing.

S. No.	Information Asset	Medium of Storage	Location	Responsibility	Acceptable Use/ Official Use (Read, Write, Email or Backup etc.)	Asset Classification	Asset Market Value (in dollars)	External Party with Access Rights
		(Electronic / Hard Copy Documents)	(Electronic Storage Location / Rack / Cabinet etc)	Owner (user/custodian)	(For the Owner)	(H, M, L)		
1	Proprietary BIM Models	Electronic	Data Centers (Beijing, London)	Architecture/Design Department Head & Managers	Read, Write, Backup	High	\$2,000,000	None
2	Client Contracts	Electronic/Hardcopy	Secure Cabinet, Encrypted Cloud Storage	Legal Team	Read, Write, Email	High	\$500,000	None
3	Research Data	Electronic	R&D Servers (Beijing, London)	R&D Team	Read, Write, Backup	High	\$1,500,000	None
4	Financial Records	Electronic	Finance Team Systems	Finance Department Team	Read, Write, Backup	High	\$1,000,000	None
5	Employee Data/Records	Electronic	HR Database	HR Department Head & Managers	Read, Write, Backup	Medium	\$200,000	None
6	IT Systems	Electronic	Regional Offices	IT Department	Operational Use	High	\$3,000,000	None
7	IoT Devices	Embedded Systems	Smart Building Infrastructure	IT Team	Monitor, Update	Medium	\$150,000	IoT Vendors
8	Data Centers	Electronic	HQ (Beijing, London)	Data Center Managers	Read, Write, Backup	High	\$5,000,000	None

Note: The Asset value of employee records actually depends on its importance. That also applies to all other assets because asset value changes from time to time.

Table 3.2: Asset Inventory Information

3. Qualitative Risk Assessment

Note: Submitted the excel sheet separately for better viewing.

No.	Asset			Exposure			
	Information Asset (Name or Description)	Asset Classification (H, M, L)	Existing Control (if any)	Threat Description	Vulnerability Description	Probability of Exposure (H, M, L)	Impact/Severity Rating (H, M, L)
0	Proprietary BIM Models	High	RBAC, Data encryption, Regular backups.	Unauthorized access	Weak access control or stolen credentials	Medium	High
				Data corruption	Insider threat or malware infection	Medium	High
2	Client Contracts	High	Data encryption, Secure cabinets, Restricted physical access.	Data Theft	Insufficient encryption or weak physical security	Medium	High
				Insider Misuse	Excessive access privileges	Medium	High
3	Research Data	High	Strong encryption, Access restrictions, Secure storage.	Data theft by competitors	Weak external-facing defenses	Medium	High
				Unauthorized access	Insider threat/Excessive privileges	Medium	High
				Corruption of files	Malware/Accidental deletion	Low	High
4	Financial Records	High	Data encryption, Secure ERP system, Compliance frameworks.	Financial fraud	Weak system controls or insider misuse	Medium	High
				Unauthorized access	Lack of RBAC and monitoring	Medium	High
				Loss of records	Inadequate backup or redundancy	Low	High
5	Employee Data/Records	Medium	Encryption, Role-based access.	Identity theft	Lack of strong encryption	Medium	High
				Unauthorized access	Poorly enforced RBAC policies	Medium	Medium
6	IT Systems	High	Endpoint Protection, Monitoring	Software Attacks (Malware Infection)	Unpatched systems or phishing attacks	Low	High
7	IoT Devices	Medium	Network segmentation, IoT-specific security protocols.	Unauthorized control	Weak passwords or insecure protocols	Medium	Medium
				Data leakage	Insufficient encryption of IoT data	Medium	Medium
				Exploited vulnerabilities	Outdated firmware	High	Medium
8	Data Centers	High	Firewalls, IDS/IPS, Redundancy measures, Disaster recovery plans.	Physical breach	Lack of robust physical security	Low	High
				Natural disasters	Insufficient disaster-resilient design	Low	High

Table 3.3: Qualitative Risk Assessment

4. Quantitative Risk Assessment

Note: Submitted the excel sheet separately for better viewing.

	Asset Name	Threat Description	Vulnerability Description	Asset Value	Exposure Rate	Single Loss Expectancy (SLE)	Annualized Rate of Occurrence (ARO)	Annual Loss Expectancy (ALE)	Applicable Control Name	Value of Control (Cost/Benefit Analysis)
1	Proprietary BIM Models	Unauthorized access	Weak access control or stolen credentials	\$2,000,000	20%	\$400,000	50%	\$200,000	RBAC, MFA	Prevents unauthorized access
		Data corruption	Insider threat or malware infection		30%	\$600,000	20%	\$120,000	Endpoint Protection	Ensures data integrity
2	Client Contracts	Data Theft	Insufficient encryption or weak physical security	\$500,000	20%	\$100,000	25%	\$25,000	Encryption, Physical Security	Protects sensitive data
		Insider Misuse	Excessive access privileges		40%	\$200,000	10%	\$20,000	RBAC, Monitoring	Reduces insider threats
3	Research Data	Data theft by competitors	Weak external-facing defenses	\$1,500,000	5%	\$75,000	10%	\$7,500	Firewalls, IDS	Prevents external theft
		Unauthorized access	Insider threat/Excessive privileges		30%	\$450,000	20%	\$90,000	RBAC, Monitoring	Mitigates insider threats
		Corruption of files	Malware/Accidental deletion		20%	\$300,000	30%	\$90,000	Backup Systems	Enables file recovery
4	Financial Records	Financial fraud	Weak system controls or insider misuse	\$1,000,000	10%	\$100,000	15%	\$15,000	Audit Trails, Authentication	Prevents financial misuse
		Unauthorized access	Lack of RBAC and monitoring		30%	\$300,000	25%	\$75,000	RBAC, MFA	Secures access control
		Loss of records	Inadequate backup or redundancy		15%	\$150,000	20%	\$30,000	Backup Systems	Ensures continuity
5	Employee Data/Records	Identity theft	Lack of strong encryption	\$200,000	20%	\$40,000	20%	\$8,000	Encryption, Monitoring	Prevents identity fraud
		Unauthorized access	Poorly enforced RBAC policies		25%	\$50,000	20%	\$10,000	RBAC Policies	Enhances access policies
6	IT Systems	Software Attacks (Malware Infection)	Unpatched systems or phishing attacks	\$3,000,000	30%	\$900,000	30%	\$270,000	Antivirus, Phishing Training	Protects against malware
7	IoT Devices	Unauthorized control	Weak passwords or insecure protocols	\$150,000	25%	\$37,500	25%	\$9,375	Strong Passwords, MFA	Secures system access
		Data leakage	Insufficient encryption of IoT data		20%	\$30,000	30%	\$9,000	IoT Encryption, Updates	Prevents IoT breaches
		Exploited vulnerabilities	Outdated firmware		20%	\$30,000	20%	\$6,000	Firmware Updates	Updates system security
8	Data Centers	Physical breach	Lack of robust physical security	\$5,000,000	10%	\$500,000	10%	\$50,000	Physical Security Systems	Secures physical assets
		Natural disasters	Insufficient disaster-resilient design		5%	\$250,000	5%	\$12,500	Backup Sites, Disaster Recovery	Ensures continuity post-disaster

Table 3.4: Quantitative Risk Assessment